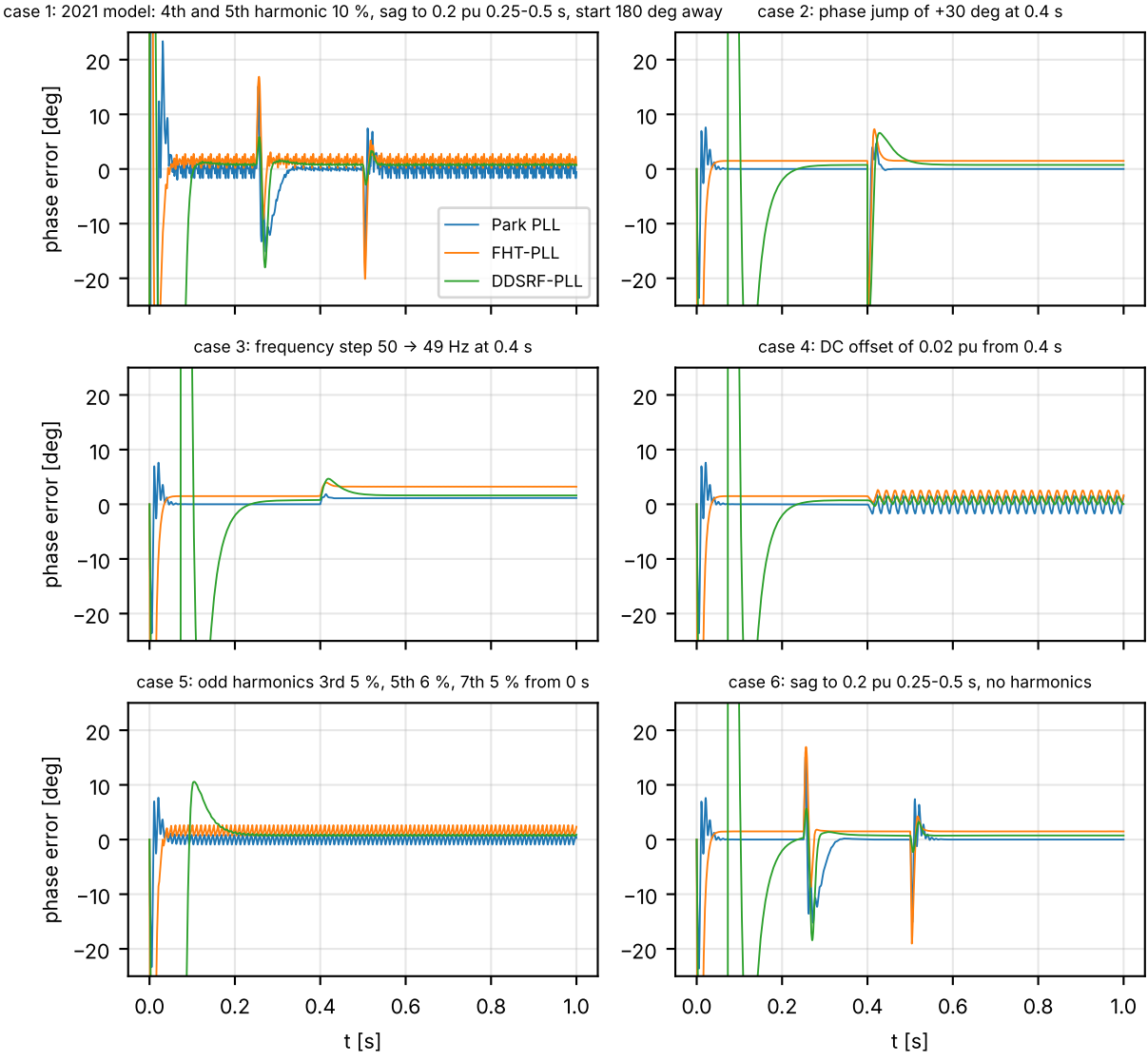

case 1: 2021 model: 4th and 5th harmonic 10 %, sag to 0.2 pu 0.25-0.5 s, start 180 deg away
case 2: phase jump of +30 deg at 0.4 s
case 3: frequency step 50 → 49 Hz at 0.4 s
case 4: DC offset of 0.02 pu from 0.4 s
case 5: odd harmonics 3rd 5 %, 5th 6 %, 7th 5 % from 0 s
case 6: sag to 0.2 pu 0.25-0.5 s, no harmonics
phase error [deg]
t [s]
Park PLL
FHT-PLL
DDSRF-PLL